

Quantitative methods

Sean Muller · October 2026

The statistical toolkit behind the research, the standards it runs under, and which checks have been applied to which backtest.

METHODS ARE DESCRIBED GENERICALLY. SIGNAL DEFINITIONS, PARAMETERS AND TRADING RULES ARE OMITTED. EVERY PERFORMANCE FIGURE IS A HISTORICAL BACKTEST.

How a result is tested

I care more about credible evidence than about preserving an attractive backtest. A question goes through eight steps before it becomes a claim:

1. **Question.** A specific question about market behavior, or about the evidence itself.
2. **Hypothesis.** What I expect to see and the economic reason for it.
3. **Design the test.** Baseline, metric, placebo and failure criteria, written before the result.
4. **Try to break it.** Placebos, dependence-aware inference, multiple-testing correction, modeled costs.
5. **Diagnose.** When it fails, find out why: the data, the method, or the idea.
6. **Revise.** Change the method or drop the claim, and record what changed.
7. **Retest.** Rerun against the same controls, on the same build and data, and count the rerun as another trial.
8. **Forward and live evidence.** The last layer: evidence that selection could not have touched.

Research standards, and the events behind them

Each rule is tied to the event that produced it, in the order the rules were adopted. The website records how each was applied afterward. One lapsed: the ledger rule was not kept in a later line of work, and the trial recount in the methodology brief is its correction.

MAY 2026 · COUNT EVENTS, NOT ROWS

A backtest passed its robustness test on a few dozen trades. A closer look at the rows found three distinct events, each counted many times. **A result is counted in independent events, not rows, and three events are too few to estimate anything.**

JUNE 2026 · LEDGER, DEFLATION, ONE HOLDOUT READ

A planned search covered hundreds of configurations against one fixed quality bar. Across that many cells, chance clears a fixed bar somewhere. **I decided that every searched cell is logged, failures included, that the bar applies only after deflating for the full count, and that the final holdout is read once.**

JUNE 2026 · SAY WHICH STAGES ARE REAL

My questions about how the backtest of an earlier line of work was built led to a stage-by-stage audit. It was a parallel re-implementation: one of six stages ran the production code, four ran simplified stand-ins and one was missing. **No backtest result is called trustworthy unless its controls pass and it comes with a list of which stages are real, stand-in or absent, and the report leads with what has not been checked.**

JUNE 2026 · PLANTED EFFECT AND NOISE FIRST

I chose a candidate statistic because it scored best on real data. Four days later a negative control on one small synthetic sample read it above chance on pure noise. **A candidate statistic is admissible only if it finds a planted effect and reads as chance on planted noise, and that is settled before its score on real data is read.**

JUNE 2026 · ATTACK THE BEST RESULT FIRST

Within a day of the best result so far, I asked for a search for errors in it by as many routes as possible. An upstream selection step had been computed on the full history, test periods included; recomputed on training data only, one of the two headline settings held and the other fell back to its baseline. **The strongest result is attacked first, and every selection step upstream of a model is held to the same point-in-time rule as the model.**

JULY 2026 · TEST ON DATA PREDATING THE RULE

Two candidate rule changes held in both halves of the history they were found on. I asked for a test on a different, longer period. It used records generated before either rule existed. One replicated and the other reversed sign. **A new rule has to hold on a different, longer period, on data generated before the rule existed; agreement between halves of the sample it was found on is not enough.**

AUGUST 2026 · REVIEW WITHOUT THE REASONING

A strong result in another line of work had passed an overfitting battery. A second review pass that I asked for saw only the data and the claim, not the reasoning behind it. It found that every control drew from one hindsight-filtered sample, and the headline did not survive. **A finding on the main strategy is not accepted until a second review pass, given only the data and the claim, has tried to refute it.**

AUGUST 2026 · REPRODUCE A KNOWN RESULT FIRST

A study reported that the strategy did not carry over to a much wider set of stocks. I did not believe an empty result across that many names. The audit found a screen applied in the wrong unit, so only about a fifth of the new stocks had been tested, and stock splits missing from the data. **A study tool has to reproduce a known result on the original stocks before any number from new territory is read.**

SEPTEMBER 2026 · SCORE NET OF MARKET EXPOSURE

On an earlier configuration, a test of shorter holding periods on a long-only book showed the shorter variants losing, and the question was closed. I objected: in a rising market such a book collects drift for as long as it holds, so a test scored on raw return can only say hold longer. **Holding-period, exit and sizing comparisons are scored on return net of fitted market exposure, with the raw figure beside it, against what the freed capital could earn.**

The toolkit

Twenty methods in six groups: the problem each answers and the method. The website adds why the naive version can fail and what each showed in this research.

EVALUATION DESIGN

Before any statistic is computed, these methods settle what a backtest can be evidence for: models fitted on earlier data and scored only on what follows, each input checked for whether it was knowable on the date it was used, and splits that keep overlapping outcomes on one side. A walk-forward test closes one leak only; the choice among configurations, the stock list, the data and the fill assumptions can each carry hindsight of their own.

Why it matters here. My first reported backtest was walk-forward for its models but read its stock list from a present-day snapshot; rebuilt as of each date, with modeled fills, the two-year result went from a superseded 4.76× to 1.87×. In the ranking feasibility study, a repeatability score of 0.48 on a random split fell to 0.02 when the same data were split by blocks of time.

Walk-forward evaluation, and what it does not protect

THE PROBLEM. A rule fitted and scored on the same data looks better than it will on new data.

THE METHOD. Walk-forward evaluation: models are fitted on earlier data, scored only on the period that follows, and rolled forward.

Point-in-time data and look-ahead tests

THE PROBLEM. A backtest can use information that was unavailable at decision time without raising an error.

THE METHOD. Point-in-time reconstruction asks of each input whether it was knowable on the date it was used. Automated tests assert that fitting data ends before each test period, and that a statistic's value on a given date is the same whether it is computed from history cut off at that date or from the full history.

Time-blocked splits and overlapping outcomes

THE PROBLEM. Trades open at the same time share part of their outcome, so they are not independent observations.

THE METHOD. Splits by blocks of time, and an effective sample size in place of the raw count: observations are discounted for overlap, and variances are inflated by a design effect measured with a bootstrap over whole days.

DEPENDENCE-AWARE INFERENCE

Daily returns and overlapping trades are not independent draws, so intervals that assume independence come out too narrow and a trade count overstates the information in the sample. These methods keep the dependence in the standard errors: block and stationary bootstraps over whole sessions, cluster-robust errors quoted with their cluster counts, and paired session-by-session differences when two backtests are compared.

Why it matters here. A rule change had been logged as the one to bring forward on the strength of two backtests; a panel of paired differences could not separate it from its reference, and the entry was corrected as premature. Where a correction was small, that is reported too: session-level and week-level intervals on the cost and execution statistics changed the sign of no conclusion.

Block bootstrap for dependent return series

THE PROBLEM. Putting an interval on a Sharpe ratio or an annual return, and a range on a drawdown, when daily returns are dependent.

THE METHOD. Block-bootstrap inference is used where temporal dependence makes iid resampling inappropriate: moving blocks of sessions, or the stationary bootstrap of Politis and Romano, with an automatic block length and longer blocks reported beside it.

Cluster-robust standard errors and serial-dependence diagnostics

THE PROBLEM. Trades entered in the same session or week share market shocks and carry less independent information than their count suggests.

THE METHOD. Diagnostics first: autocorrelation of daily returns and of absolute returns, the share of overlapping trades, and the intraclass correlation of outcomes within a session. Cluster bootstraps that resample whole sessions or whole weeks, and cluster-robust standard errors, one-way and two-way. Every clustered t-statistic is quoted with its cluster count. Rows with few clusters are labeled unreliable, not corrected.

Comparing two backtests: paired differences and compounding noise

THE PROBLEM. The gain from a rule change is a difference between two backtests, and a compounded ending multiple is path-dependent.

THE METHOD. The paired difference between candidate and reference, session by session, with a dependence-aware interval, so noise common to both arms cancels. For the multiple's noise, the same rules rerun from neighboring start dates, and the spread between runs that differ only by an arbitrary change or a placebo.

MULTIPLE TESTING AND SELECTION BIAS

When many variants are tried on one sample, the best of them is biased upward by the selection, and any correction needs an honest count of what was tried. These methods count the trials and estimate how many were effectively independent, test the best candidate jointly with all the others against a benchmark fixed in advance, and place a selected Sharpe ratio on a probability scale against the best that luck alone would produce.

Why it matters here. My first reported backtest deflated its Sharpe ratio for an assumed 100 trials; a recount of the record found about 540 runs in 62 hypothesis families, and the claim was restated as a range. The October 2026 sweep was then judged once, as one family of 69 candidates beside 44 placebo twins: one survived on the whole sample, an effect already known, and it did not hold in the later half of the sample taken alone.

Trial accounting and the effective number of independent trials

THE PROBLEM. A multiplicity correction needs a count of trials. Backtests of related variants are highly correlated, so neither the raw count nor one is right.

THE METHOD. An inventory of every recorded run, grouped into hypothesis families, and an effective number of independent trials estimated several ways from the correlations between runs' daily returns, reported as a range.

Family-wise tests for the best of many candidates

THE PROBLEM. Deciding whether the best of many candidates on one sample beats a benchmark or is only the luckiest.

THE METHOD. White's Reality Check and Hansen's test of superior predictive ability bootstrap all candidates jointly, in blocks, against a benchmark fixed in advance. The Romano–Wolf step-down gives each an adjusted p-value.

Probabilistic and deflated Sharpe ratio

THE PROBLEM. A Sharpe ratio selected from many backtests on a short, fat-tailed sample is overstated.

THE METHOD. The probabilistic Sharpe ratio places an observed Sharpe ratio on a probability scale against a reference value, allowing for sample length, skewness and kurtosis. Bailey and López de Prado's deflated version sets that reference to the best Sharpe expected by luck among the trials run, given how widely their Sharpe ratios vary. It is not a reduced Sharpe ratio, and it is not the probability that the strategy has skill. It is one minus the one-sided p-value for the hypothesis that the true Sharpe is no higher than the reference.

PLACEBO CONTROLS, POWER AND FEASIBILITY

A passing test means little unless it could have failed, and a null means little unless the test could have found an effect of the size that matters. These methods supply both sides: placebo and negative controls that keep the mechanics of a pipeline and remove its information, planted effects of known size to measure what the sample can detect, and a noise ceiling that says what a perfect model could reach before another model is built.

Why it matters here. In the October 2026 sweep the 44 placebo twins gave no rejections, as a null family should, while planted effects of two realistic sizes were found only about a fifth and about a third of the time, so most nulls in that sweep mean an effect of that size could not have been seen. Measuring the noise ceiling then closed a line of work as below resolution: the true spread between signals was more than ten times smaller than the noise in one outcome.

Placebo and negative controls, including placebo twins

THE PROBLEM. A pipeline can report an effect that comes from its own mechanics, not from information.

THE METHOD. Controls that keep the mechanics and remove the information: shuffled labels, scrambled inputs, a random draw in place of a rule's information. Placebo twins of the candidates enter the same family-wise test and must fail.

Planted-effect power analysis and the minimum detectable effect

THE PROBLEM. A null means little unless the test could have found an effect of the size that matters.

THE METHOD. Effects of known size are planted into simulated draws that keep the candidates' measured correlations and standard errors. Each decision rule is applied and detections are counted. The smallest effect the sample could detect is stated.

The noise ceiling: what a perfect model could achieve

THE PROBLEM. Knowing whether a prediction task is achievable with the data available, before building models.

THE METHOD. Split-half reliability on time-blocked halves estimates how much of the variation in outcomes is repeatable. The ratio of that repeatable spread to single-outcome noise sets the rank correlation a ranker that knew each group's true expected value would reach. It is a ceiling for the groupings and the outcome examined, not for every possible input.

Learned models judged against placebo twins

THE PROBLEM. A flexible learner may find structure that hand-written rules miss, or fit pure noise. Its scores have to order outcomes and add to a simple baseline.

THE METHOD. Gradient-boosted trees, a pairwise ranker and regularized linear models, tested as candidate rankers with nested chronological tuning, and scored by Spearman rank correlation with realized outcomes and by partial rank correlation controlling for a simple baseline. Each has placebo twins: the same recipe fitted on scrambled inputs, shuffled labels or pure noise.

RETURN, RISK AND ATTRIBUTION STATISTICS

A backtest reduces to a few comparable numbers only if each carries its convention: how the Sharpe ratio is annualized, which curve the drawdown is measured on, and how skewness and fat tails enter the sampling error. These methods also ask where the profit came from: how concentrated it is in a few trades, and how much of it is exposure to the market and to known factors rather than anything specific to the strategy.

Why it matters here. The concentration measures produced a limit stated beside the headline: a small fraction of trades accounts for more than half of net profit. Factor and benchmark decomposition, together with leave-out diagnostics, continue as an evidence-extension layer for characterizing where historical performance originates.

Return and tail statistics, and their conventions

THE PROBLEM. Reducing a backtest to a few comparable numbers when daily returns are skewed and fat-tailed.

THE METHOD. The Sharpe ratio is the annualized ratio of mean daily return to daily volatility, with no risk-free rate subtracted, so it reads higher than an excess-return Sharpe ratio when rates are positive. Skewness and kurtosis of daily returns enter its sampling error. Drawdown is measured on the daily marked curve. Tail risk is historical value at risk and expected shortfall. On about two years of daily returns, both rest on a small number of tail days.

Concentration of profit

THE PROBLEM. A result can be carried by a handful of trades, and that decides how it should be described.

THE METHOD. The share of net profit from the top fraction of trades, the share of trades needed to reach half of it, the Gini coefficient and Lorenz curve of winning trades' profit, and a Herfindahl index of absolute profit shares. Leave-out tests recompute the result without the largest winners.

Factor attribution by time-series regression

THE PROBLEM. A long-only book in a rising market earns market return by construction. Its profit may be known-factor exposure, not anything specific to the strategy.

THE METHOD. Ordinary least squares of daily account returns on the market, then on standard factor portfolios from the academic literature and sector indexes, with Newey–West standard errors. It reports the share of profit the factors explain and the uncertainty on the remainder.

EXECUTION AND COST MODELING

A flat cost with every order filled at the signal price overstates what a real order earns, so cost is modeled as a function of spread, liquidity, volatility, order size, market impact and fill uncertainty. These methods also quantify uncertainty in the cost calibration, decompose the gap between decision-price profit and executable profit, and test whether a more complex order schedule actually improves held-out outcomes.

decompose the gap between decision-price profit and executable profit into its causes, and test whether a cleverer order schedule beats a simple one.

Why it matters here. Replacing the flat cost with modeled fills lowered the backtest result, and modeled cost now rises with the simulated starting balance; a price gain seen only under the most favorable fill assumption was not claimed. A formal optimal-control schedule and learned policies were tested against fixed schedules and placebo twins, and none beat the best simple schedule on held-out data, so the simulation keeps the simple rule.

Transaction-cost model and its fill assumptions

THE PROBLEM. A flat cost with every order filled in full at the signal price overstates what a real order earns.

THE METHOD. Trading cost is modeled as a function of spread, liquidity, volatility and order size rather than as a constant per-trade penalty. Market impact and fill uncertainty are allowed to vary with trading conditions, and the cost and fill assumptions are stressed under alternative conventions, including deliberately adverse cases. The calibration remains an uncertain model input rather than a fixed fact.

Estimating the impact coefficient

THE PROBLEM. The impact coefficient sets how fast cost grows with order size. It is the model's weakest input.

THE METHOD. Two regression estimators on public trade-and-volume data, with intervals from resampled sessions. Each is combined with a prior from the academic literature in a one-parameter Bayesian update, its likelihood widened because the data are a proxy.

Implementation shortfall decomposition

THE PROBLEM. Profit at the decision price is not profit that can be executed. The gap has several causes, each with a different remedy.

THE METHOD. Perold's implementation shortfall, per trade on both legs. Profit at the decision price, less delay, half-spread, impact, fees and the opportunity cost of unfilled shares, equals executable profit, and the identity closes exactly.

Whether a cleverer schedule beats a simple one: optimal control and learned policies

THE PROBLEM. Trading faster costs more impact. Trading slower risks the price moving away or an incomplete order.

THE METHOD. The schedule is posed as a stochastic control problem with temporary market impact and a liquidity constraint: in continuous time its value function satisfies a Hamilton–Jacobi–Bellman equation; here it is solved in discrete time as a Bellman recursion, by backward induction, and checked against closed forms, including the Almgren and Chriss linear-impact case. The learned alternative, fitted Q-iteration and contextual bandits, is tested against fixed schedules and placebo twins.

What has been applied to which backtest

This table shows which checks have been applied to which backtest. Run means the check was carried out, not that the result passed it; Planned means the check is scheduled for that backtest and is not a result; the repeats now in progress are directed first at the selected configuration.

CHECK	FIRST REPORTED BACKTEST (ARCHIVED)	ORIGINAL CONFIGURATION, RESTATED	SELECTED CONFIGURATION
Point-in-time universe	No	Run in part	Run in part
Modeled fills	No	Run	Run
Cost by starting balance	Not applicable	Run	Run
Deflated Sharpe ratio	Run	Planned	Run
Dependence-aware intervals on the Sharpe ratio	Run	Planned	Planned
Start-date sensitivity	Run	Planned	Planned
Leave-out tests for the largest winners	Run	Planned	Planned
Factor attribution	Run	Planned	Planned
Execution and fill-stress studies	Run	Planned	Planned
Research sweep and its family-wise test	Reference for the sweep	Planned	Planned
Placebo comparison for the rule revision	Not applicable	Not applicable	Run

The first reported backtest is the most examined and is archived. The two point-in-time columns are the most recent, and the selected configuration is where the planned checks are now directed. No means the backtest was built without it: the first reported backtest used a present-day stock list and a flat cost, and the restated column is where those two were corrected. Run in part refers to the point-in-time reconstruction, which is still being completed. Not applicable means the row does not describe that backtest, and Reference for the sweep means that backtest served as the sweep's benchmark rather than being tested by it.

The research questions, methodology, system design, and interpretation are independently directed; implementation uses extensive AI-assisted development.

Sean Muller · seanmuller988@gmail.com · [linkedin.com/in/seancmuller](https://www.linkedin.com/in/seancmuller) · github.com/seanmuller988-ui · seanmuller988-ui.github.io

Selected material presented for educational and professional purposes. Proprietary signal definitions, parameters, portfolio construction rules and execution logic are omitted. All performance figures are historical backtests; no live or paper-trading results are shown. Historical and simulated results do not guarantee future performance. Not an offer or investment advice.